

A sharp Gaussian first-moment inequality for every prescribed mass vector

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Abstract

We establish a sharp first-moment inequality for every measurable simplex-valued function on Gaussian space with prescribed positive mean. The label metric is the negative Hessian of the Gaussian multi-bubble perimeter profile. The equality functions are exactly the indicators of the translated regular simplex partition with those probabilities, extended cylindrically. The result holds for every number of labels and every ambient dimension. Its proof derives a global tangent bound for the squared optimal Gaussian assignment value from the established multi-bubble perimeter theorem and a radial differential inequality. An exact nonnegative integral measures its deficit. At equal masses this recovers the sharp ordinary squared-centroid sum; at unequal masses the metric is generally anisotropic. We do not assert the false unequal-mass Euclidean simplex conjecture or the full functional multi-bubble isoperimetric inequality. The profile Hessian identity is prior work; our statement is a bound for all functions, not only perimeter minimizers.

1. Model profile and the sharp functional theorem

Let $k \geq 2$, $n = k - 1$, $\mathbf{1} = (1, \dots, 1)^T$, $P = I - \mathbf{1}\mathbf{1}^T/k$, and $Q_* = P/n$. Fix centered rows v_i^* of a $k \times n$ matrix V_* with $V_*V_*^T = Q_*$. Their pairwise distances are $\ell_* = \sqrt{2/n}$.

Fix a probability vector $p = (p_1, \dots, p_k)$ with every $p_i > 0$. There are unique prices λ_p in the gauge $\sum_i \lambda_{p,i} = 0$ such that

$$\Omega_i(p) = \{x : v_i^* \cdot x - \lambda_{p,i} \geq v_j^* \cdot x - \lambda_{p,j} \ \forall j\} \quad (1)$$

has standard Gaussian probability p_i in $\mathbb{R}^n$. Their existence, uniqueness and smooth dependence are proved in Section 2. The score matrix maps $\mathbb{R}^n$ onto $\mathbf{1}^\perp$, so $\lambda_p = V_* w_p$ for a unique w_p . Thus (1) is a central regular simplex fan translated by w_p .

Write A_{ij}^p for the ordinary Gaussian interface areas of this model. Define two label Laplacians, annihilating $\mathbf{1}$:

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$$(H_p u)_i = \sum_{j \neq i} A_{ij}^p (u_i - u_j), \quad L_p = H_p / \ell_*.$$

All model facets have positive area, so both matrices are positive on $\mathbf{1}^\perp$. The notation H_p^+, L_p^+ means inversion on this subspace and zero on the constant line. Put

$$I(p) = \sum_{i < j} A_{ij}^p, \quad c_p = \frac{\text{tr } L_p}{n} = \ell_* I(p) > 0. \quad (2)$$

By Milman–Neeman [MN22, Theorem 1.1], I is the Gaussian k -cell perimeter profile in dimension $n = k - 1$. Perimeter is normalized as half the sum of individual cell perimeters, or equivalently the sum of ordinary pairwise interfaces. Its Hessian on the tangent space $\mathbf{1}^\perp$ satisfies the established identity

$$\nabla^2 I(p) = -H_p^+. \quad (3)$$

This identity is [MN22, Proposition 2.4]; Section 2 also derives its normalization from the price calculus.

Theorem 1. Let $d \geq 1$ and let $f_i : \mathbb{R}^d \rightarrow [0, 1]$ be measurable, with $\sum_i f_i = 1$ almost everywhere and $\int f_i d\gamma_d = p_i$. Let the $k \times d$ moment matrix B have rows $b_i = \int x f_i(x) d\gamma_d(x)$. Then

$$\text{tr}(B^T H_p^+ B) \leq I(p), \quad (4)$$

or equivalently

$$-\sum_{a=1}^d B_{\cdot,a}^T \nabla^2 I(p) B_{\cdot,a} \leq I(p). \quad (5)$$

If $d \geq k - 1$, equality holds exactly when the f_i are the ordinary winning indicators (1) in an orthogonally embedded $(k - 1)$ -dimensional subspace, with its orthogonal complement unrestricted, up to null sets and label symmetries preserving the fixed mass vector. If $d < k - 1$, the inequality is strict for every such function; no sharp fixed-lower-dimensional bound is claimed.

This is a universal functional first-moment bound. It applies to all measurable partitions and fractional labels of mean p , rather than only to variations of perimeter minimizers. The profile identity (3) and the perimeter theorem are explicitly imported prior results. The finite literature check [LS] found no equivalent all-function statement in the checked sources; it does not establish worldwide novelty.

2. Prescribed-mass Gaussian score calculus

For every symmetric positive-semidefinite covariance Q with $Q\mathbf{1} = 0$, define

$$\mathcal{C}_p(Q) = \inf_{\lambda \in \mathbb{R}^k} \left\{ p \cdot \lambda + \mathbb{E} \max_i (Y_i - \lambda_i) \right\}, \quad Y \sim N(0, Q). \quad (6)$$

Adding a common price has no effect. In the gauge $\min \lambda_i = 0$, the expression is at least $p_{\min} \max \lambda_i$: a zero-priced mean-zero score makes the expected maximum nonnegative. Hence the infimum is attained. Pointwise comparison with the p -weighted average of scores also shows $\mathcal{C}_p(Q) \geq 0$.

For distinct score rows v_i , Gaussian pairwise ties are null. The price gradient is $p_i - \gamma_n(C_i)$, where $C_i = \{x : v_i \cdot x - \lambda_i \geq v_j \cdot x - \lambda_j \ \forall j\}$. Minimizing prices therefore give precisely the masses p_i . Their winning assignment attains the score maximum pointwise. For any feasible ordinary or fractional labels f with mean p , and any centered score matrix M , the same pointwise comparison at every price gives

$$\langle M, B \rangle \leq \mathcal{C}_p(MM^T). \quad (7)$$

No assignment interpretation at duplicated scores is needed.

On the entire closed covariance cone,

$$\mathcal{C}_p(sQ) = \sqrt{s} \mathcal{C}_p(Q) \quad (s \geq 0), \quad \mathcal{C}_p \text{ is continuous.} \quad (8)$$

Scale scores and prices to prove homogeneity, and note $\mathcal{C}_p(0) = 0$. For continuity couple $Q^{1/2}G$ and $\tilde{Q}^{1/2}G$. The difference of their maxima at any price is at most $|(Q^{1/2} - \tilde{Q}^{1/2})G|$. Its expectation bounds the two infima uniformly in prices. Continuity of the matrix square root therefore covers singular endpoints as well.

Suppose Q is positive on $\mathbf{1}^\perp$, and write $Q = MM^T$ with centered rows in $\mathbb{R}^n$. They are affinely independent. The n score differences against one label give nonsingular affine coordinates. They realize a top tie between any chosen pair with all remaining scores strictly lower. Consequently every pair facet has positive Gaussian area. Define its length $\ell_{ij} = |v_i - v_j|$, area A_{ij} , weight $w_{ij} = A_{ij}/\ell_{ij}$, and weighted Laplacian L .

Gaussian integration by parts, with inward facet normal $(v_i - v_j)/\ell_{ij}$, gives the winning-cell moments $B(M) = LM$. The additional boundary terms after truncation with balls tend to zero exponentially. Since the winning masses are p , price terms cancel, so

$$\mathcal{C}_p(Q) = \text{tr}(LQ) = \sum_{i < j} w_{ij} \ell_{ij}^2, \quad \text{tr } L = 2 \sum_{i < j} w_{ij}. \quad (9)$$

Lemma 2. On this positive cone $\mathcal{C}_p$ is smooth and

$$d\mathcal{C}_p(Q)[D] = \frac{1}{2} \text{tr}(LD) \quad (10)$$

for every symmetric centered covariance direction D .

Proof. In the affine score-difference coordinates, winning regions are fixed polyhedral cones. Gaussian density and linear-integrand derivatives are locally dominated by a polynomial times a Gaussian, so these integrals are smooth. The price Hessian is L : a price gap moves the pair facet by that gap divided by ℓ_{ij} . All pair weights are positive, making L positive on the zero-sum gauge at every price. Thus balancing prices are unique in that gauge, and the implicit function theorem makes them smooth.

Write $D = MAM^T$ with A symmetric; this representation exists because M maps onto $\mathbf{1}^\perp$. Realize its covariance path by $M(t) = M(I + tA)^{1/2}$. The envelope score derivative is $B(M)$ and $M'(0) = MA/2$. Formula (9) then gives $d\mathcal{C}_p(Q)[D] = \text{tr}(B(M)^T MA)/2 = \text{tr}(LD)/2$. $\square$

This proves the assertions about model prices in Section 1. In the model $\ell_{ij} = \ell_*$; (9) gives $\mathcal{C}_p(Q_*) = \ell_* I(p) = \text{tr } L_p/n = c_p$, including all factors in (2). For completeness the derivative of the mass vector with respect to model prices is $-L_p$ on the tangent gauge. The envelope derivative of c_p with respect to p is λ_p . Therefore $\nabla^2 c_p = -L_p^+$, and $\nabla^2 I = -L_p^+/\ell_* = -H_p^+$, as in the prior identity (3).

3. Perimeter and a global covariance tangent bound

Let Q be positive on $\mathbf{1}^\perp$ and of trace one. Its balanced winning cells have masses p in dimension $n = k - 1$, precisely the range $k \leq n + 1$ of the imported multi-bubble theorem. Write their perimeter $S = \sum_{i < j} A_{ij}$. Thus $S \geq I(p)$. Weighted Cauchy–Schwarz and (9) give

$$S^2 = \left(\sum_{i < j} w_{ij} \ell_{ij} \right)^2 \leq \left(\sum_{i < j} w_{ij} \right) \left(\sum_{i < j} w_{ij} \ell_{ij}^2 \right) = \frac{1}{2} \mathcal{C}_p(Q) \text{tr } L.$$

Using $I(p) = c_p \sqrt{n/2}$, we obtain

$$\mathcal{C}_p(Q) \frac{\text{tr } L}{n} \geq c_p^2. \quad (11)$$

Every weight is positive, so equality in weighted Cauchy requires all pair lengths equal. A centered trace-one score Gram with this property is Q_* : double-centering its squared-distance matrix gives $Q = (\ell^2/2)P$, and trace one gives $\ell^2 = 2/n$.

Theorem 3. For every centered positive-semidefinite covariance Q ,

$$\mathcal{C}_p(Q)^2 \leq c_p \text{tr}(L_p Q). \quad (12)$$

Equality holds exactly when $Q = sQ_*$ for some $s \geq 0$.

Proof. First normalize $\text{tr } Q = 1$, and put $Q_t = Q_* + t(Q - Q_*)$ for $0 \leq t < 1$. This is positive on $\mathbf{1}^\perp$ even when Q is singular. Write $C(t) = \mathcal{C}_p(Q_t)$ and L_t for its facet matrix. Formulas (9)–(10) imply

$$tC'(t) = \frac{1}{2} (C(t) - \text{tr } L_t/n). \quad (13)$$

Let $h(t) = C(t)^2 - c_p^2$ and $D_p(t) = C(t) \text{tr } L_t/n - c_p^2 \geq 0$. Then exactly

$$(h(t)/t)' = -D_p(t)/t^2 \leq 0. \quad (14)$$

The initial derivative need not vanish. By (10) its value is

$$\lim_{t \downarrow 0} h(t)/t = h'(0) = c_p \text{tr}(L_p(Q - Q_*)). \quad (15)$$

At one the quotient tends to $\mathcal{C}_p(Q)^2 - c_p^2$ by (8). Integrating (14) on compact subintervals and taking both endpoint limits therefore gives the exact identity

$$c_p \operatorname{tr}(L_p Q) - \mathcal{C}_p(Q)^2 = \int_0^1 \frac{D_p(t)}{t^2} dt \geq 0. \quad (16)$$

This nonnegative improper integral is finite because both quotient limits are finite. Smoothness near zero also gives $D_p(t) = O(t^2)$. Here $c_p \operatorname{tr}(L_p Q_*) = c_p^2$ was used to simplify the endpoint difference. This proves (12) at trace one.

If equality holds, continuity and nonnegativity make $D_p(t) = 0$ for every $0 < t < 1$. The perimeter/Cauchy chain consists of equalities, and all pair lengths are equal, forcing $Q_t = Q_*$. Any such $t > 0$ implies $Q = Q_*$. Conversely the model has equality. For nonzero Q of trace $s > 0$, apply the result to Q/s and use homogeneity (8). At $Q = 0$ both sides vanish. This gives all stated equality rays, including singular endpoints. $\square$

Writing $W(t) = \sum_{i < j} w_{ij}(t)$, the integrand has the decomposition

$$D_p(t) = \frac{2}{n} \left[W(t) \sum_{i < j} w_{ij}(t) \ell_{ij}(t)^2 - S(t)^2 + S(t)^2 - I(p)^2 \right]. \quad (17)$$

Both terms are nonnegative: weighted edge-length variance and squared-perimeter deficit. No metric stability rate or full covariance concavity is asserted.

4. Moment duality, sharpness and all equality cases

For the functions in Theorem 1 the rows of B sum to zero. Put $E = \operatorname{tr}(B^T L_p^+ B) \geq 0$ and $M = L_p^+ B$. The feasible score inequality (7) and Theorem 3 imply

$$E = \langle M, B \rangle \leq \mathcal{C}_p(MM^T), \quad \mathcal{C}_p(MM^T)^2 \leq c_p \operatorname{tr}(L_p MM^T) = c_p E. \quad (18)$$

The last equality uses $L_p^+ L_p L_p^+ = L_p^+$. Thus $E^2 \leq c_p E$, giving $E \leq c_p$. Since $H_p = \ell_* L_p$ and $c_p = \ell_* I(p)$, this is exactly (4), and the tangent-space identity (3) gives (5).

If $E = c_p > 0$, all comparisons in (18) are equalities and $\mathcal{C}_p(MM^T) = c_p$. Theorem 3 gives $MM^T = sQ_*$; homogeneity then gives $\sqrt{s}c_p = c_p$, so $s = 1$. Thus M is a trace-one regular simplex score list. Equality in the feasible assignment means its nonnegative pointwise maximum deficit has zero integral. Its scores have null ties, so the fractional labels must be the unique winning indicators. Uniqueness of the prescribed-mass prices identifies exactly (1) after an orthogonal embedding. This requires $d \geq n$; below that dimension equality is impossible. If $E = 0$ the inequality is strict.

Conversely model flux gives $B_* = L_p V_*$, whence

$$\operatorname{tr}(B_*^T L_p^+ B_*) = \operatorname{tr}(V_*^T L_p V_*) = \operatorname{tr}(L_p Q_*) = c_p.$$

Orthogonal and cylindrical extensions preserve moments and masses. This proves Theorem 1, its sharpness and every equality assertion. The proof includes $k = 2$, where the normalized covariance space is a singleton. The omitted $k = 1$ case has one cell and zero moments. No statement at a zero prescribed probability is needed.

5. Equal masses and the scope of the result

At $p_i = 1/k$, symmetry gives $L_p = c_k P$, where $c_k = \mathbb{E} \max_{i \leq k} Z_i / \sqrt{k-1}$ for independent standard normals. Theorem 1 then reduces to

$$\sum_i |b_i|^2 \leq \frac{(\mathbb{E} \max_{i \leq k} Z_i)^2}{k-1}.$$

This recovers [EQ], which proves the equal-mass subcase of [H19, Conjecture 1.16] and the full four-cell dimension-three question [H14, Conjecture 3]. At unequal masses, L_p is generally not scalar on label space; the nonzero initial slope (15) supplies new metric information that the uniform statement does not contain. The universal bound (4) cannot follow merely by calibrating its quadratic form on the model moments; (11)–(18) prove it for all feasible functions.

The ordinary unweighted prescribed-mass squared-moment conjecture is false [EX]. The present theorem uses the explicitly defined label metric H_p^+ and makes no contrary Euclidean assertion. It is a first-moment functional inequality, not the full simplex-valued Gaussian Bobkov or positive-correlation noise-stability statement. The profile Hessian identity is prior work; its application here bounds all functions, beyond variations of perimeter minimizers.

6. Proof and verification dependencies

The non-elementary geometric input is [MN22, Theorem 1.1], used only with $k = n + 1$, its positive-mass hypotheses and its interface-perimeter normalization. The established profile Hessian identity is stated with its tangent-space convention and independently normalized in Section 2. The remaining tools are Gaussian integration by parts, dominated convergence, the implicit function theorem, matrix square roots, pseudoinverses and one-variable calculus. There is no numerical or solver premise, no differentiation at a singular endpoint, no assumption of covariance concavity and no rank classification.

The package includes internal model reviews, finite exact diagnostics and partial Lean proofs with explicit scope. They do not certify Gaussian measure, the imported geometric theorem or the entire analytic endpoint. Internal model reviews are not external human peer review. The earlier equal-mass result is preserved as its own immutable record; this theorem is a generalization of the same proof mechanism, not an independent counting of the equal-mass case.

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